Zijian (Jack) Luo

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Education

The University of Sydney — Mar 2024 – Mar 2026
MPhil in Computer Science — *Sydney, Australia*

Hebei University — Sep 2018 – Jun 2022
B.Eng. in Software Engineering — *Hebei, China*

Work Experience

Microsoft Research Asia (MSRA) — Mar 2026 – Present
Research Intern — *Beijing, China*

- Enhanced **Copilot's cross-platform code migration** capability by integrating a **knowledge base** to model platform-specific APIs, dependencies, and migration mappings.
- Built a **benchmark suite** to evaluate Copilot on **long-horizon software engineering tasks** measuring end-to-end planning and execution across multi-step workflows for **agentic RL**.

Oracle Labs, Oracle Corporation — May 2025 – Oct 2025
Research Assistant — *Brisbane, Australia*

- Conducted research on **software supply-chain reliability** by integrating LLMs with the Macaron framework, reproducing build artifacts for over **120 open-source repositories** from Libraries.io with an **80% success rate**.
- Designed LLM-assisted workflows for defect reproduction and **static-analysis** validation.

Publications

- **Automatic Data Repair without Format Specifications.**
  Zijian Luo, Lukas Kirschner, Ezekiel Soremekun, Rahul Gopinath.
  IEEE International Symposium on Software Reliability Engineering (ISSRE) 2025.

- **Assessing Reliability of Statistical Maximum Coverage Estimators in Fuzzing.**
  Danushka Liyanage*, Nelum Attanayake*, **Zijian Luo***, Rahul Gopinath.
  IEEE International Conference on Software Maintenance and Evolution (ICSME) 2025.

- **Maximal Format-Free Record Repair.**
  Zijian Luo, Hong Jin Kang, Alan Fekete, Rahul Gopinath.
  Under Blinded Review. A black-box repair framework that infers intrinsic program input structure using regular-language and automata learning, enabling correction without format specifications.

Selected Projects

- **The Deeplang Programming Language (82 Stars)**
  A memory-safe, multi-paradigm programming language for embedded systems. I was responsible for implementing the backend virtual machine, including interpreting and executing WebAssembly (WASM) code generated by the frontend.

Technical Skills

Programming Languages	C/C++, Python, Rust, WebAssembly
Analysis & Testing	Fuzzing (AFL), Program Repair, Grammar Inference, CodeQL
AI	LLMs, Agentic RL